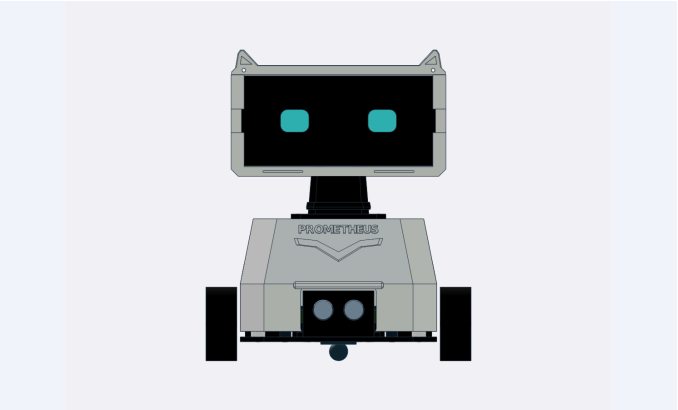
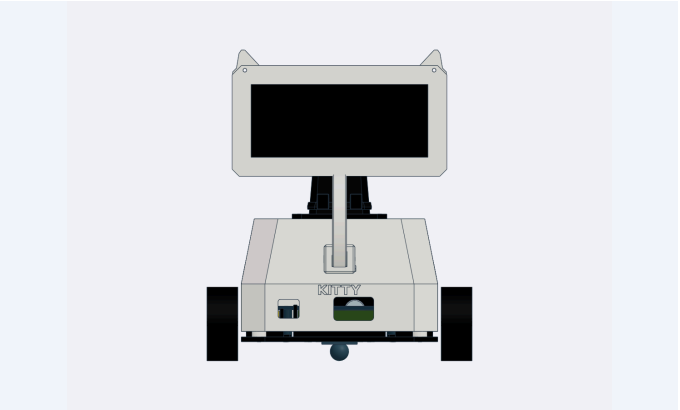
KITTY V7 / Team PROMETHEUS

Print-service order summary · 24 October 2026 (IST)

> **PRINT THE FIT STAGE FIRST — PHYSICAL FIT IS NOT APPROVED**





First order: five pieces

File	Qty	Purpose
v7_fit_tail_socket.stl	1	0.3 / 0.4 / 0.5 mm clearance per side
v7_fit_tail_key.stl	1	Matching key section and insertion
v7_fit_phone_thickness.stl	1	8.8 mm thickness throat; not full phone fit
v7_fit_screws.stl	1	3.2 / 3.4 / 3.6 / 3.8 mm holes
v7_motor_left.stl	1	Real motor/shaft/tie and fastener dry fit; retain if good

The four samples are in FIT_TEST_FIRST/. The left cradle is in STL_PARTS/ and counts toward the body quantity. Use actual motors, wheels, phone/case and fasteners.

Confirm before ordering large parts

- Motor reservation: 65 × 24 × 20 mm plus proposed clearance, from user-estimated ranges. Slots offer ±2 mm fore/aft adjustment.
- Wheel outside diameter: 62.5 mm. Axial width/hub and shaft height are unmeasured; the model's 9 mm shaft-centre assumption needs checking with the caster and wheel engagement.
- Phone report: 164 × 76 × 8 mm; check case, buttons, camera, cable and retention. AA holder is unbought and planned outside; its external restraint is unresolved.
- Current production tail gap is 0.4 mm per side; holes are 3.4 mm. Update CAD after coupon results if needed. Do not scale an entire STL to repair a fit.

Production and repair

STL_PARTS/ contains 13 separate body files; the internal battery tray is optional. No wheel, motor, phone, electronics or whole-robot STL is supplied. Use millimetres at 100%. PETG/FDM is a quote option; service-specific profile, orientation, supports and strength remain unverified.

Use removable screws/ties. Check actual M3 head/washer fit; model checks cover 8 mm OD × 0.8 mm washers and 5.5 mm OD × 3 mm heads. Small modules can be replaced individually. A broken integral tail socket still requires a rear-cover replacement.

Read PRINT-SERVICE-BRIEF.md, PRINT-REVIEW.md and REPAIR-MAP.csv. Digital solid/mesh/interference checks passed; no physical fit, printing or load test has been performed. This summary is not a dimensioned manufacturing drawing or a printer profile.